

Smart Room Automation - Step-by-Step Build Guide

Arduino Uno + PIR + LDR + TMP36 | Tinkercad build walkthrough (as followed)

Ye guide poore build process ko wahi order mein follow karta hai jaise aap ne step by step kiya — har step complete kar ke agle par jayein.

Step 1: Naya Circuit Banayein

- 1 Browser mein tinkercad.com kholein aur login karein.
- 2 Left menu mein "Circuits" par click karein.
- 3 "Create new Circuit" button dabayein (top right).
- 4 Ek naya blank workplane khulega — ye aapka canvas hai.

Step 2: Arduino Uno Board Add Karein

- 1 Right side Components panel ke search bar mein type karein: "Arduino Uno R3".
- 2 Result ko click-and-drag kar ke workplane ke left side mein rakhein.
- 3 Thodi khali jagah chhod dein right side pe (breadboard ke liye).
- 4 USB port left ki taraf rakhein — isse wiring baad mein aasan rahegi.

Step 3: Breadboard Add Karein

- 1 Search bar mein type karein: "Breadboard" (ya "Small Breadboard").
- 2 Drag kar ke Arduino ke right side mein rakhein.
- 3 Arduino aur breadboard ke darmiyan thodi space rakhein.
- 4 Breadboard ko horizontally align kar lein (rows seedhi rahein).

Step 4: PIR Sensor, LDR, aur TMP36 Add Karein

- 1 "PIR sensor" search karein → "Motion Sensor (PIR)" ko breadboard ke neeche/side mein rakhein.
- 2 "Photoresistor" (LDR) search karein → breadboard ke upar left side rakhein (do legs, do rows).
- 3 "TMP36" search karein → LDR ke thodi door right side rakhein (teen legs, teen rows).

Zaroori baat: har component ko un rows mein rakhein jahan legs seedhi holes mein fit ho jayein — Tinkercad green highlight se confirm karta hai.

Step 5: LED, Resistors, aur Buzzer Add Karein

- 1 "LED" search karein, koi bhi color choose karein, TMP36 ke right side rakhein (lambi leg = anode +, choti leg = cathode -).
- 2 "Resistor" do baar add karein: ek LED ki cathode side, doosra LDR ke sath (voltage divider ke liye).
- 3 LED wala resistor click kar ke value 220 ohm set karein.
- 4 LDR wala resistor click kar ke value 10k (10000) ohm set karein.
- 5 "Buzzer" search karein, LED ke right side rakhein.

Step 6: Power Rails Wire Karein (5V aur GND)

- 1 Arduino ke 5V pin se wire nikaal kar breadboard ki + rail mein lagayein. Color: red.
- 2 Arduino ke GND pin se wire nikaal kar breadboard ki - rail mein lagayein. Color: black.

Ye power backbone ready karta hai jisse baad mein har component VCC/GND le sakta hai.

Step 7: LDR Wire Karein (Light Sensor)

- 1 LDR ki ek leg se wire nikaal kar breadboard ki + rail (5V) mein lagayein. Color: red.
- 2 LDR ki doosri leg ko 10k ohm resistor ke ek sirre ke sath same row mein connect karein.
- 3 Usi row se wire nikaal kar Arduino ke A0 pin mein lagayein. Color: blue.
- 4 Resistor ka doosra sirra breadboard ki - rail (GND) mein connect karein. Color: black.

Voltage divider logic: LDR upar se 5V leta hai, neeche resistor se GND jata hai, beech ka point A0 ko signal deta hai.

Step 8: TMP36 Wire Karein (Power / Vout / Ground)

Power (Vs / +Vs) - Red wire

From	To
TMP36 Power pin (left leg)	Breadboard + rail (5V)

Vout (signal) - Blue wire

From	To
TMP36 Vout pin (middle leg)	Arduino A1

Ground - Black wire

From	To
TMP36 Ground pin (right leg)	Breadboard - rail (GND)

Yaad rakhne ka tarika: TMP36 ka flat (printed) side apni taraf face karein — left-se-right order hamesha Power → Vout → Ground hota hai. Middle pin hamesha Vout hoti hai.

Step 9: LED Wire Karein (Room Light)

- 1 LED ki lambi leg (anode, +) se wire nikaal kar Arduino ke D9 pin (~PWM) mein lagayein. Color: orange.
- 2 LED ki choti leg (cathode, -) ko 220 ohm resistor ke sirre ke sath same row mein connect karein.
- 3 Resistor ka doosra sirra breadboard ki - rail (GND) mein connect karein. Color: black.

LED ka current hamesha resistor se ho kar GND mein jana chahiye — direct GND mein nahi, warna LED simulation mein error de sakti hai.

Step 10: PIR Sensor aur Buzzer Wire Karein

Buzzer

- 1 Buzzer + pin se wire nikaal kar Arduino ke D8 pin mein lagayein. Color: green.
- 2 Buzzer - pin se wire nikaal kar breadboard ki - rail (GND) mein lagayein. Color: black.

PIR Sensor - Power

From	To
PIR VCC pin	Breadboard + rail (5V)

PIR Sensor - Ground

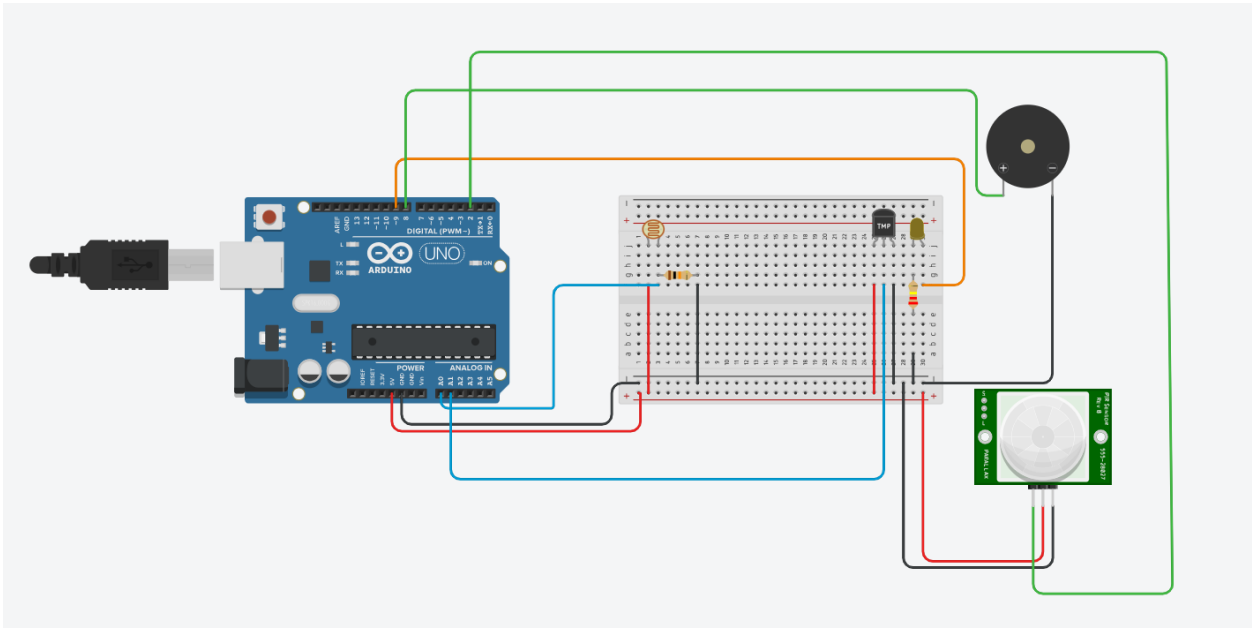
From	To
PIR GND pin	Breadboard - rail (GND)

PIR Sensor - Signal

From	To
PIR OUT pin	Arduino D2

Pin order yaad rakhne ka tarika: PIR par teen pins hoti hain, aksar left-se-right order GND — OUT — VCC hota hai (Tinkercad mein hover kar ke label zaroor confirm karein).

Circuit Diagram (Breadboard View)



As-built breadboard layout: Arduino Uno + Breadboard + PIR Sensor + LDR + TMP36 + LED + Buzzer.

Step 11: Code Paste Karein aur Simulate Karein

- 1 Tinkercad workplane ke top right mein "Code" button par click karein.
- 2 Dropdown se "Blocks" se "Text" mode select karein.
- 3 Agar default code ho to pura delete kar dein.
- 4 Neeche wala poora sketch copy-paste karein.
- 5 "Start Simulation" button dabayein (top right, green).
- 6 PIR par click kar ke motion simulate karein, LDR par click kar ke light level adjust karein, TMP36 par click kar ke temperature 30°C se upar le jayein aur buzzer test karein.

Full Arduino Sketch

```
// ===== Smart Room Automation =====
// PIR (motion) + LDR (light level) + TMP36 (temperature)
// -> auto room light (LED, PWM) + temperature alert (buzzer)

const int pirPin    = 2;    // PIR OUT (digital)
const int buzzerPin = 8;    // Buzzer +
const int ledPin    = 9;    // LED anode (PWM)
const int ldrPin    = A0;   // LDR voltage divider midpoint
const int tempPin   = A1;   // TMP36 signal

const int darkThreshold = 500;    // below this = "dark" (0-1023 scale)
const float tempAlertC  = 30.0;   // buzzer triggers above this temp

void setup() {
  pinMode(pirPin, INPUT);
  pinMode(buzzerPin, OUTPUT);
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int motion = digitalRead(pirPin);
  int ldrValue = analogRead(ldrPin);

  int tempReading = analogRead(tempPin);
  float voltage = tempReading * (5.0 / 1023.0);
  float temperatureC = (voltage - 0.5) * 100.0;    // TMP36 formula

  // --- Room light logic ---
  bool isDark = ldrValue < darkThreshold;
  if (motion == HIGH && isDark) {
    int brightness = map(ldrValue, 0, darkThreshold, 255, 80);
    analogWrite(ledPin, brightness);
  } else {
    analogWrite(ledPin, 0);
  }

  // --- Temperature alert logic ---
  if (temperatureC > tempAlertC) {
    digitalWrite(buzzerPin, HIGH);
  } else {
    digitalWrite(buzzerPin, LOW);
  }

  // --- Debug output ---
  Serial.print("Motion: "); Serial.print(motion);
  Serial.print(" | LDR: "); Serial.print(ldrValue);
  Serial.print(" | Temp: "); Serial.print(temperatureC);
```

```
Serial.println(" C");  
  
delay(300);  
}
```

Simulation ke baad agar koi error ya unexpected behavior aaye, screenshot le kar dobara check karwaya ja sakta hai.